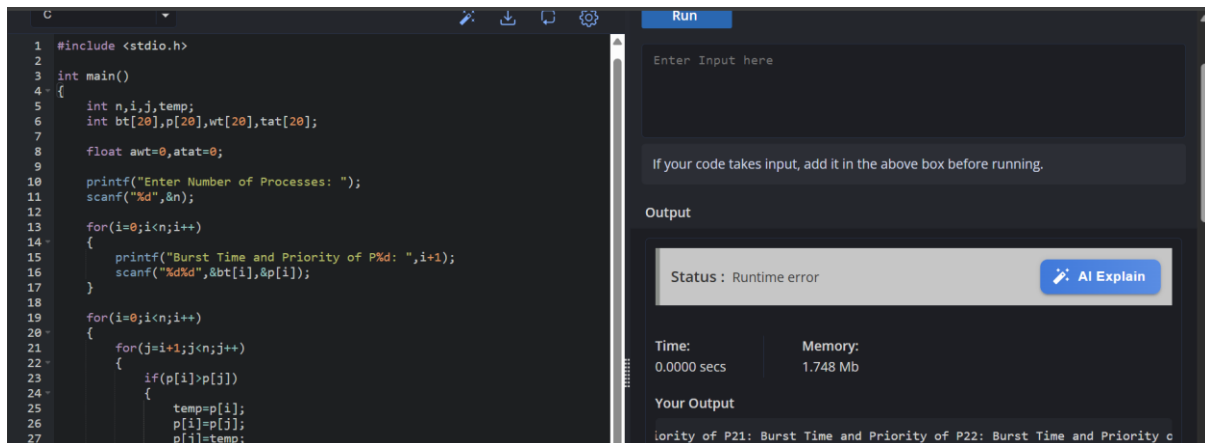


EXPERIMENT 05:

5. Construct a scheduling program with C that selects the waiting process with the highest priority to execute next.



The screenshot shows a C program in a code editor and its execution output. The code implements a scheduling algorithm that reads the number of processes, their burst times, and priorities, then sorts them by burst time. The output shows a runtime error.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n,i,j,temp;
6     int bt[20],p[20],wt[20],tat[20];
7
8     float awt=0,atat=0;
9
10    printf("Enter Number of Processes: ");
11    scanf("%d",&n);
12
13    for(i=0;i<n;i++)
14    {
15        printf("Burst Time and Priority of P%d: ",i+1);
16        scanf("%d%d",&bt[i],&p[i]);
17    }
18
19    for(i=0;i<n;i++)
20    {
21        for(j=i+1;j<n;j++)
22        {
23            if(p[i]>p[j])
24            {
25                temp=bt[i];
26                bt[i]=bt[j];
27                bt[j]=temp;
```

Run

Enter Input here

If your code takes input, add it in the above box before running.

Output

Status : Runtime error [AI Explain](#)

Time: 0.0000 secs Memory: 1.748 Mb

Your Output

iority of P21: Burst Time and Priority of P22: Burst Time and Priority c

PSEUDO CODE :

Start

Input number of processes

Input Burst Time

Sort processes by Burst Time

Waiting Time of first process = 0

For remaining processes

Waiting Time = Sum of previous burst times

Turnaround Time = Waiting Time + Burst Time

Calculate averages

Display results

Start

Input number of processes

Input Burst Time

Input Priority

Sort by Priority

Waiting Time of first process = 0

Calculate Waiting Time

Calculate Turnaround Time

Calculate averages

Display results

Stop

SAMPLE OUTPUT :

Process BT Priority WT TAT

P2 3 1 0 3

P1 5 2 3 8

P3 4 3 8 12

Average Waiting Time = 3.67

Average Turnaround Time = 7.67

CODE:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,i,j,temp;
```

```
    int bt[20],p[20],wt[20],tat[20];
```

```
    float awt=0,atat=0;
```

```
    printf("Enter Number of Processes: ");
```

```
    scanf("%d",&n);
```

```
    for(i=0;i<n;i++)
```

```
    {
```

```
        printf("Burst Time and Priority of P%d: ",i+1);
```

```
        scanf("%d%d",&bt[i],&p[i]);
```

```
    }
```

```
    for(i=0;i<n;i++)
```

```
    {
```

```
        for(j=i+1;j<n;j++)
```

```
        {
```

```
            if(p[i]>p[j])
```

```
            {
```

```
                temp=p[i];
```

```

        p[i]=p[j];

        p[j]=temp;

        temp=bt[i];

        bt[i]=bt[j];

        bt[j]=temp;
    }
}

wt[0]=0;

for(i=1;i<n;i++)

    wt[i]=wt[i-1]+bt[i-1];

for(i=0;i<n;i++)
{
    tat[i]=wt[i]+bt[i];

    awt+=wt[i];

    atat+=tat[i];
}

printf("\nBT\tPriority\tWT\tTAT\n");

for(i=0;i<n;i++)

    printf("%d\t%d\t\t%d\t\t%d\n",bt[i],p[i],wt[i],tat[i]);

printf("\nAverage Waiting Time = %.2f",awt/n);

printf("\nAverage Turnaround Time = %.2f",atat/n);

return 0;
}

```